Fine-Grained Computation Offload for Event-Driven Applications

Bojie Li
Pine AI

Abstract

Hardware accelerators now sit on the critical path of online serving: GPUs, TPUs, FPGAs, and increasingly *remote* services such as hardware security modules, post-quantum cryptography, and inference servers. While the accelerator works, the calling context stalls. For *fine-grained* offloads (microseconds to a few milliseconds), the classic responses both fall short: a context switch adds a wakeup latency comparable to the offload, and a busy-wait wastes the core. The principled fix is to *overlap* the offload with other useful work, but this requires injecting concurrency into the application. The real question is *how much the application must change*.

We study this across real, off-the-shelf servers and find a *spectrum*. At one extreme, an `LD_PRELOAD` M:N fiber runtime overlaps the offload with *zero* source modification, but works only for thread-per-connection servers and hits three architectural walls: synchronization below libc, thread identity in native thread-local storage, and per-request CPU that already exceeds the offload. At the other extreme, a *few lines* at the offload site route it through the application's own concurrency and work everywhere. Across ten production servers spanning every concurrency model the change is **18–112 lines added and 0–1 modified**, and real overlap lands at 1.2–5× on stock servers (17.3× for the transparent runtime), all measured on real hardware. A simple model predicts both the speedup and the code cost from two properties: the server's concurrency model and the offload's weight relative to per-request work, and the win is bounded at both ends, by the accelerator's throughput and by the server's own overlap capacity. A transparent page-protection conflict detector preserves correctness when the overlapped offload mutates shared state, validated on stock Redis.

Code: https://github.com/19PINE-AI/transparent-offload
Website: https://01.me/research/transparent-offload

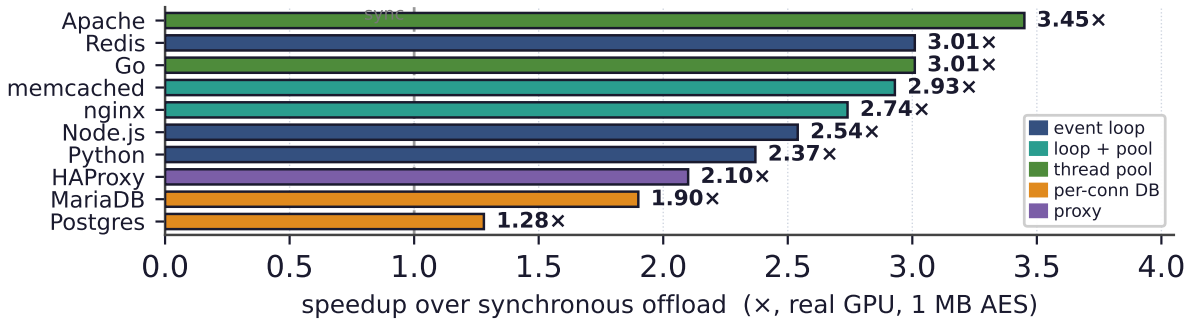

Figure 1. The headline result. A few lines of overlap recover 1.2–3.5× over a synchronous offload across ten stock servers (real GPU, 1 MB AES); the transparent runtime reaches 17.3× with zero edits. Colour marks the concurrency model.

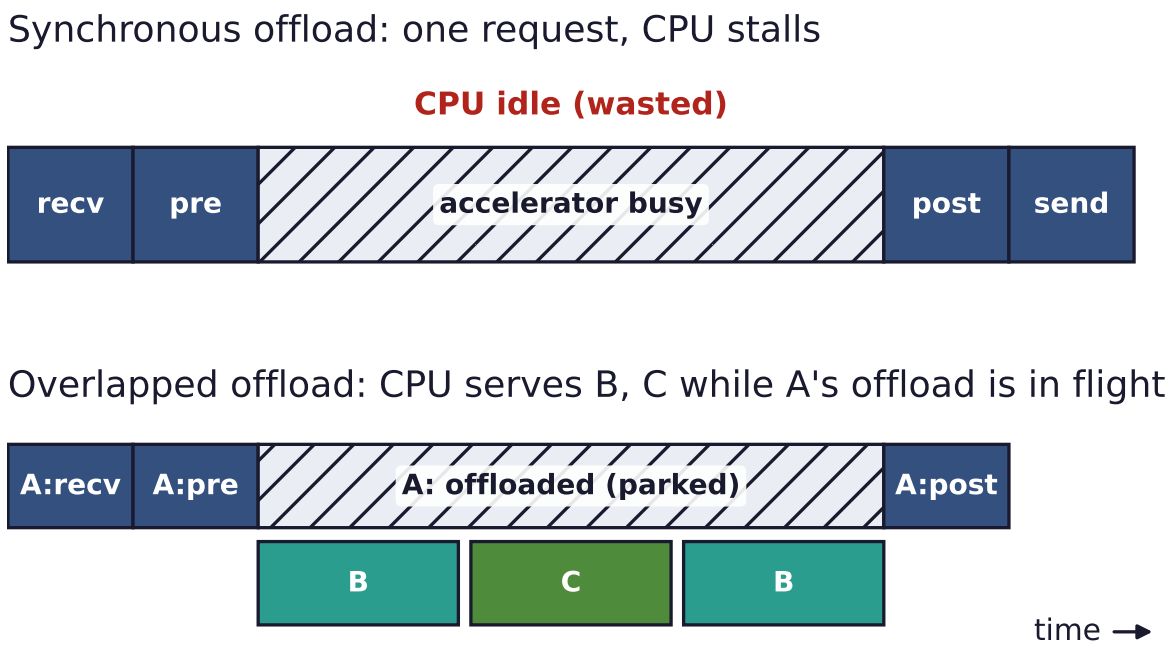

Figure 2. The problem. A serving request is *receive* → *pre-process* → *offload* → *post-process* → *send*. Top: with a synchronous offload, the CPU stalls while the accelerator works. For fine-grained offloads (microseconds to a few milliseconds), a context switch to fill the gap costs as much as the gap itself, and busy-waiting wastes the core. Bottom: *overlap* fills the gap with other requests' processing. The question of this paper is how much an existing server must change to do this.

1 Introduction

Hardware accelerators are now a standard part of online serving. GPUs, TPUs, and FPGAs accelerate machine-learning inference, image processing, encryption, and compression, and an increasing share of "acceleration" is in fact a *remote* call: to a hardware security module that signs a token, to a post-quantum key-encapsulation service National Institute of Standards and Technology [2024], or to a co-located inference server Crankshaw et al. [2017]. The shape of the application is remarkably uniform across these cases. A server receives a request from the network, does some pre-processing, invokes a computationally intensive routine, does some post-processing, and sends a response. When the intensive routine runs on an accelerator, it is replaced by an *offload*: a submission to the device and a wait for its completion. This raises a deceptively simple question that this paper is about: *what should the CPU do while it waits for the accelerator?*

The textbook answers are unsatisfying precisely in the regime that matters (Figure 2). The first is to *relinquish the CPU*: after submitting the offload, block, and let the operating system run another thread. But a context switch on Linux costs several microseconds, and a fine-grained offload, whether AES on a few kilobytes, a small inference, or an HSM round-trip, also takes from a few to tens of microseconds. This is the "killer microsecond" regime Barroso et al. [2017]: too long to spin away, too short for the operating system to schedule around profitably. The switch overhead is then comparable to the work it overlaps. It wastes CPU and *adds* a thread-wakeup delay to the request's tail latency. The second answer is to *busy-wait* on the completion flag, which hides the wakeup delay but burns a core doing nothing. The third, and the only principled one, is to *do other useful work* on the same CPU while the offload is

in flight. This means *overlapping* the offload with the processing of other requests. It is what high-performance systems already do by hand. The difficulty is that overlapping requires *concurrency* inside the application at the offload point, and existing servers express concurrency in many different ways. The central question of this paper is therefore not *whether* to overlap, but *how much an off-the-shelf server must be changed to do so*.

We answer this empirically, across the landscape of real servers, and the answer is a **spectrum** bounded by two extremes (Figure 9). At one extreme, overlap can be injected with *no source modification at all*. We build a user-space M:N fiber runtime, loaded by `LD_PRELOAD`, that interposes the standard threading and I/O symbols: it turns each connection-handling thread into a lightweight fiber, yields the fiber at every blocking point, and runs other fibers on the same core in the meantime. On a synchronous thread-per-connection handler it delivers large speedups, 17.3× over a synchronous baseline for GPU AES, with the application binary unchanged. But transparency reaches only so far, and mapping exactly how far is itself a contribution. Running the runtime under stock binaries surfaces a sequence of obstacles that any threading-interposing tool must solve (most notably a glibc condition-variable symbol-versioning hazard that silently corrupts some binaries). Beyond those lie *three architectural walls* that no engineering removes: storage engines that synchronize with raw `futex` system calls *below* the libc interface, managed runtimes that store thread identity in a native thread-local slot the loader cannot virtualize, and workloads whose per-request CPU already exceeds the offload so that there is no idle time to reclaim. Crucially, the runtime engages only on *thread-per-connection* servers. An event-driven server has a single event loop and no per-connection thread to fiberize. So the very class of applications with the *most* to gain from overlap, where one synchronous offload stalls the *entire* server, is the class transparency cannot help.

At the other extreme, overlap can be injected with *a few lines*. The recipe is uniform: replace the synchronous offload at its call site with an asynchronous one that hands the work to the application's *own* concurrency mechanism, such as a module's background thread pool, an event loop's deferred-response primitive, or a language runtime's tasks. The request is then answered on completion. The edit is small because every modern server already contains machinery to suspend and resume work. One only has to route the offload through it. We instantiate this recipe on **ten** off-the-shelf servers spanning every concurrency model: Redis, nginx, memcached, Node.js, Python/asyncio, Apache, Go, PostgreSQL, MariaDB, and HAProxy. In each case the change is **18 to 112 lines added, and zero or one existing line modified**. Measured on real hardware throughout, with no emulated latencies, the edit recovers 1.2–5× with no failed requests (Figure 1). It is bounded at both ends, by the accelerator's throughput and by how much concurrency the server can itself express (§7).

Behind these numbers is a simple predictive model, the paper's main intellectual contribution. Two properties of a server fix both the speedup *and* the code cost of overlapping an offload: its *concurrency model*, which places it in a regime that predicts how much a minimal edit buys and how many lines it takes (dominated by whether the server already exposes a deferred-response primitive), and the offload's *weight relative to per-request CPU*, which predicts whether overlap helps at all. The model explains the full range of our measurements, including where overlap does *not* help, and tells an operator in advance which servers to touch and what to expect.

Overlap also introduces one correctness hazard. When the post-processing after an offload mutates shared state, overlapping two requests can reorder their read-modify-write sequences

and corrupt it. We provide a transparent conflict detector, based on page protection and version snapshots, that flags exactly these conflicts and serializes only the conflicting work, at both ends of the spectrum and with no application annotation.

In summary, this paper makes the following contributions:

- **A predictive model** of fine-grained offload overlap in existing servers: speedup and code cost as a function of the server's concurrency model and the offload's weight relative to per-request work, validated across ten real servers (§5).
- **The transparent boundary, characterized**: a zero-modification M:N fiber runtime, the obstacle taxonomy it must overcome (including a reusable glibc condition-variable versioning bug), three architectural walls where transparency provably stops, and a classifier that predicts whether a binary is fiberizable (§3).
- **A minimal-edit recipe and a cross-landscape study**: one offload-overlap pattern on ten off-the-shelf servers with 18–112 lines added and 0–1 lines modified, evaluated on real hardware (1.2–5×). We also find that the win is bounded at both ends: by the accelerator's throughput (a large GPU offload saturates the device's bandwidth) and by the server's own overlap capacity (a real latency-bound remote signer tops out at 3.5× behind a GIL-bound executor) (§4, §7).
- **A transparent conflict detector** that keeps overlapped offload correct when it mutates shared state, with no application changes (§6).

2 Background and Motivation

Accelerators and the fine-grained regime. An accelerator turns a CPU-bound routine into a device operation: the CPU prepares an input buffer, submits it (a kernel launch, a DMA, or a network send), and later collects the result. These round-trips span microseconds to milliseconds (Figure 3), but what unites them is that they are *fine-grained* relative to operating-system scheduling: the offload finishes on a timescale comparable to a context switch. There the classic ways of waiting are wasteful (Figure 2)—blocking pays a switch and a wakeup that rival the offload, busy-waiting burns a core—so overlapping the offload with other work is the only approach that neither wastes the CPU nor inflates latency.

How servers express concurrency. "Other work to overlap with" means *concurrent requests*, and a server's ability to overlap an offload is determined by how it runs concurrent requests in the first place. We find it useful to group servers into a few models, which recur throughout the paper. A *single-event-loop* server (e.g., Redis, Node.js, a Python `asyncio` service) multiplexes all connections on one thread that loops over ready events. An *event-loop-with-pool* server (nginx, memcached) runs a few such loops plus a worker pool for blocking work. A *thread- or goroutine-pool* server (Apache, Go) dispatches each request to a pooled worker or a runtime-scheduled task. A *proxy* (HAProxy) forwards each request through a native offload engine rather than running application logic itself. A *process- or thread-per-connection* server (PostgreSQL, MariaDB) gives each connection its own backend. These models differ sharply in what a synchronous offload *costs* and in how overlap must be injected, differences this paper turns into a predictive model (§5).

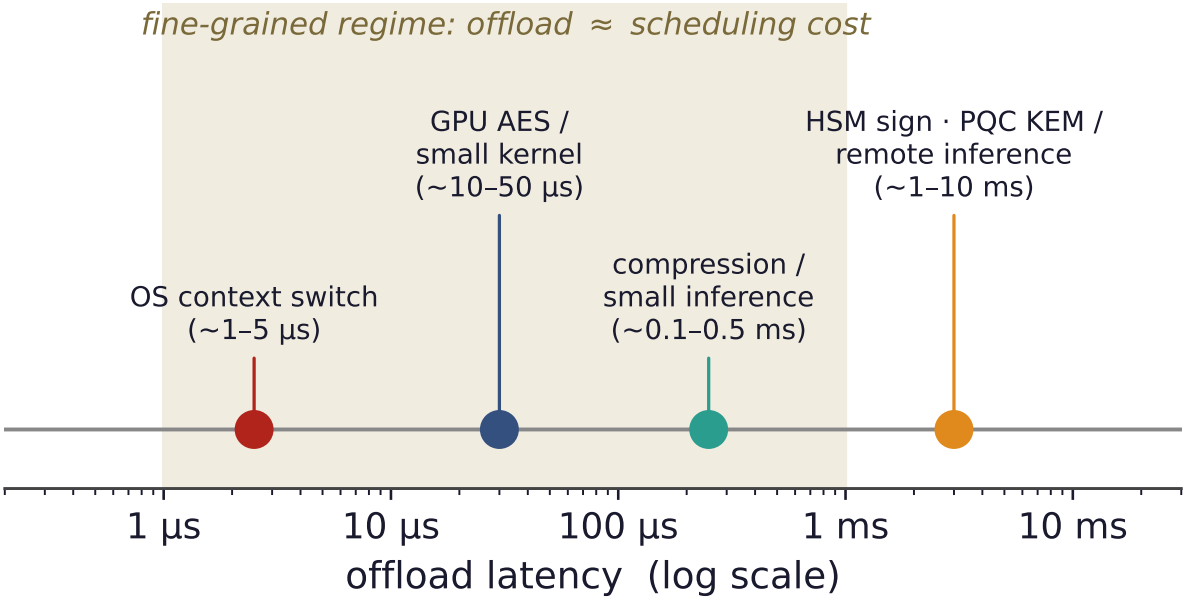

Figure 3. The fine-grained regime. Accelerator offloads span microseconds to milliseconds. In the shaded band, the offload latency is comparable to an OS context switch, so blocking pays a switch and a wakeup that rival the work itself, and busy-waiting wastes a core. This is exactly where overlapping the offload with other requests is the right answer.

The motivating catastrophe. The starkest case is a single event loop: because one thread handles every connection, a synchronous offload blocks *all* connections for its full duration, and throughput collapses to one request per offload latency while the hardware sits idle—a pathology of *structure*, not capacity. A few lines that move the offload off the loop recover more than an order of magnitude (§7). The event-driven servers that dominate modern serving thus have the most to gain from overlap and, as we show next, are also the ones a transparent runtime cannot help.

3 Transparent Overlap: The Zero-Edit Extreme

We begin at the zero-modification end of the spectrum. The goal is the most convenient form of overlap imaginable: take an *unmodified* server binary and make it overlap its offloads, with no recompilation and no source access. The result is an existence proof that the mechanism works. More importantly for the rest of the paper, it is a precise map of how far transparency can reach.

3.1 An `LD_PRELOAD` M:N fiber runtime

Our runtime (Figure 4) is a shared library loaded ahead of the application by `LD_PRELOAD`. It interposes the standard threading and I/O symbols. When the application creates a connection-handling thread, the runtime instead spawns a *fiber*: a user-level thread with its own small stack and a hand-written, register-only context switch of tens of nanoseconds, some twenty times cheaper than a kernel switch Li et al. [2023]. All fibers run on a single *carrier* OS thread. When a fiber would block, whether on a socket read or write or on the offload, the runtime saves its context and switches to another runnable fiber. A scheduler on the carrier waits for

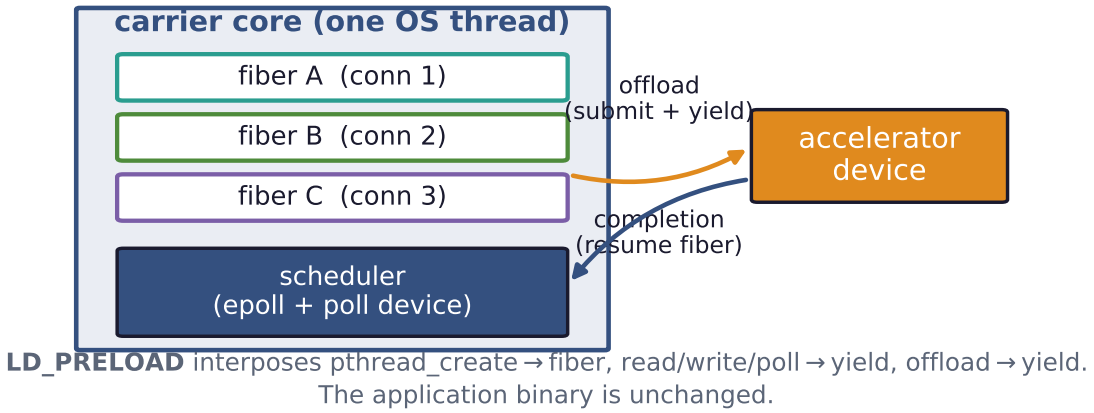

Figure 4. The transparent runtime. An `LD_PRELOAD` library turns each connection thread into a fiber on one carrier core. A fiber yields at its offload (and at socket I/O), and the scheduler runs another fiber until the device completes. The application binary is unchanged. On a synchronous thread-per-connection handler with a real GPU this overlaps 64 connections' offloads and delivers 17.3× over a blocking baseline.

I/O readiness and offload completions and resumes the corresponding fiber. The application's handler keeps its plain, synchronous shape: read a request, call the offload, write a reply. It never learns that its "thread" is a fiber or that its offload was overlapped with its neighbors'. Per-fiber state that the C library treats as thread-local, such as `errno` and the OpenSSL error queue, is saved and restored at every switch so that fibers do not see each other's transient state.

On a synchronous thread-per-connection handler this works well. With a real GPU performing AES, the runtime overlaps 64 connections' offloads on one carrier core and reaches 17.3× the throughput of the same binary blocking on each offload, with results verified bit-for-bit. A DNN-inference server shows 11.8×. These results confirm the mechanism. The rest of this section is about its limits, which turn out to be the more useful contribution.

3.2 What it takes to run real binaries

Interposing the threading layer of a stock binary is harder than it sounds. The sharpest obstacle is a *symbol-versioning* hazard Drepper [2011]: the glibc condition-variable functions carry two incompatible ABIs under one name, so a naive interposer that defines an *unversioned* `pthread_cond_signal` breaks the linker's version-matched relocation and silently corrupts some binaries (MariaDB crashes at startup with a wild pointer) while sparing others (Figure 5). The fix is to export the interposed symbols at their exact version and resolve the real ones with `dlvsym`. The defect is latent in any threading-interposing tool, a reusable practitioner lesson. Several lesser obstacles—alternate I/O entry points, real-thread identity for libc's stack-bounds logic, carrier re-establishment after a daemon's `fork`, and priority inversion under real-time pinning—we resolved once each; Appendix B records them.

3.3 Three walls where transparency stops

Beyond engineering obstacles lie three *walls* that no amount of interposition removes (Figure 6). Each is a place where the application's behavior escapes the layer a preloaded library can

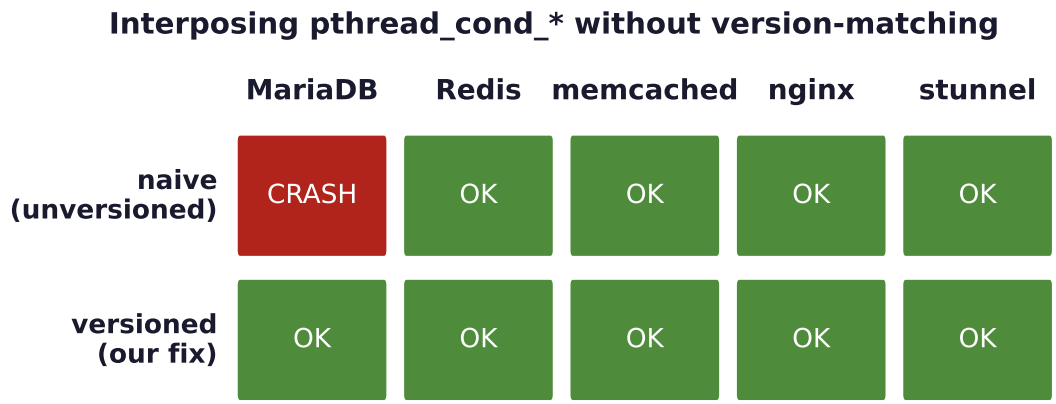

Figure 5. A reusable interposition hazard. The glibc condition-variable functions carry two incompatible ABIs under one name. A naive unversioned interposer breaks the version-matched relocation and corrupts *some* binaries (MariaDB crashes at startup) while silently sparing others. Version-matching the interposed symbols fixes all of them. Any threading-interposing tool must do this.

intercept.

Wall 1: synchronization below libc. A storage engine such as InnoDB does its internal locking with the `futex` system call Franke et al. [2002] issued *directly*, bypassing the `pthread` layer entirely. A userspace runtime interposes library *symbols*, not system calls. A fiber that blocks in a raw `futex` therefore stalls the entire carrier, and under contention the whole server deadlocks. We confirmed this with a live backtrace of the frozen carrier inside InnoDB's synchronization code.

Wall 2: thread identity in native TLS. A managed runtime such as the JVM stores "the current thread" in a native thread-local slot read directly from a CPU register, not through any library call. Fibers that share one carrier OS thread cannot be given distinct values of that slot by symbol interposition, so after a fiber switch the runtime reads the *wrong* thread object and segfaults. We observe exactly this: a crash dereferencing a stale `Thread*` the moment two fiberized JVM requests interleave. The same mechanism applies to Go and .NET.

Wall 3: no idle CPU to reclaim. Overlap only helps if the CPU is idle during the offload. A TLS terminator that is already thread-per-connection and spends real CPU on per-request crypto has no such idle time: the OS already overlaps its offloads across threads, while the single carrier serializes the real crypto and pays an interposition tax on every yield. The result is 2–3× *slower* than native on the same core.

3.4 A classifier for fiberizability

The walls are not arbitrary. They are predictable from a binary's behavior. We profile the *per-request blocking syscalls* a server makes under load. Event-driven servers do their I/O with `epoll` and `read` and have no per-connection thread to fiberize: the runtime loads safely but never engages. Thread-per-connection servers whose blocking is all at the libc layer have *near-zero* per-request `futex` traffic and are fiberizable. Engines that synchronize below libc have `futex` traffic that dominates their profile and use kernel-bypass I/O such as `io_uring` Axboe [2019]. This is the wall. Figure 7 makes it concrete: InnoDB's per-request `futex` density is roughly two orders of magnitude above a fiberizable server's, a cheap static-plus-dynamic

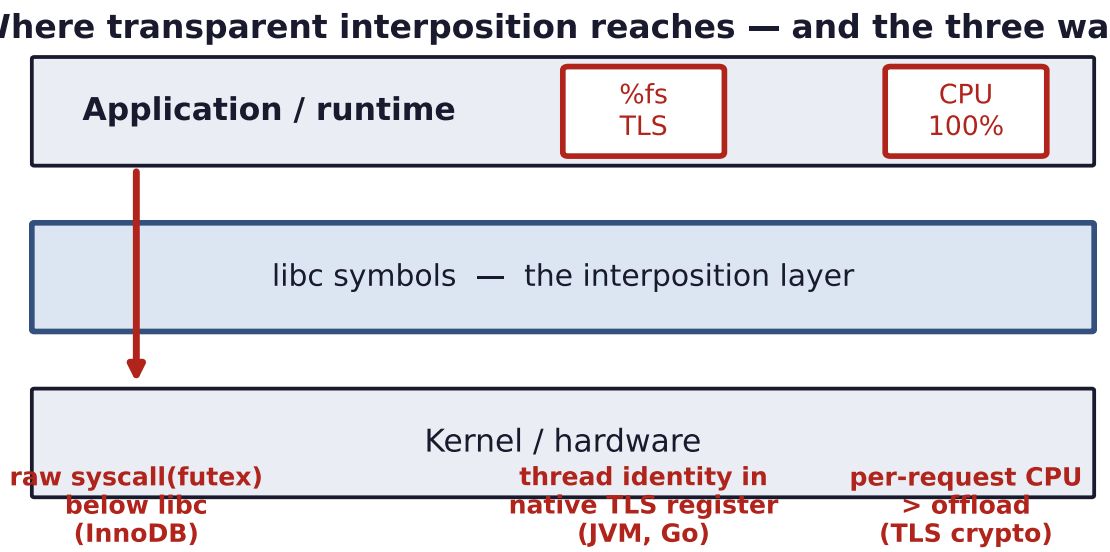

Figure 6. The three walls. Transparent interposition virtualizes the libc symbol layer (read/write, the `pthread` primitives, `errno`). It cannot reach three things. The first is synchronization issued as a *raw* system call below libc (InnoDB's `futex`). The second is thread identity held in a native TLS register (the JVM, Go). The third is a workload where per-request CPU already exceeds the offload, leaving no idle time. These bound where zero-edit overlap is possible.

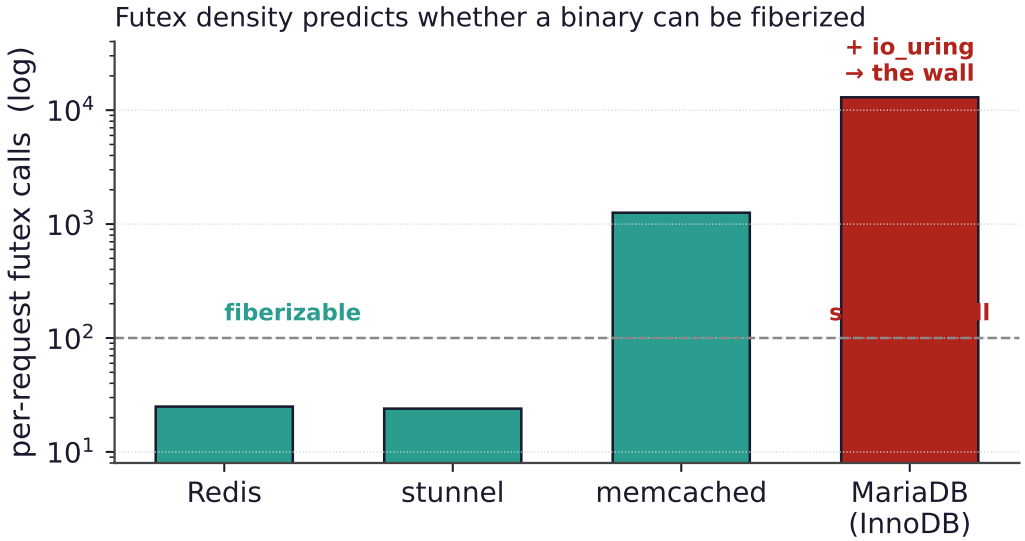

Figure 7. Predicting fiberizability. Per-request `futex` calls separate the near-zero libc-level servers (whose blocking the runtime can interpose) from engines that synchronize below libc (InnoDB, ∼13,000, plus `io_uring`), the wall of Figure 6. The same profile flags event-driven servers as "loads safely, no overlap."

check that tells an operator in advance whether a binary can be fiberized.

The verdict frames the rest of the paper. Transparency serves a narrow niche: thread-per-connection servers with light per-request CPU and a libc-level offload. By construction, it cannot help the event-driven servers that have the most to gain. For everyone else, a few lines suffice.

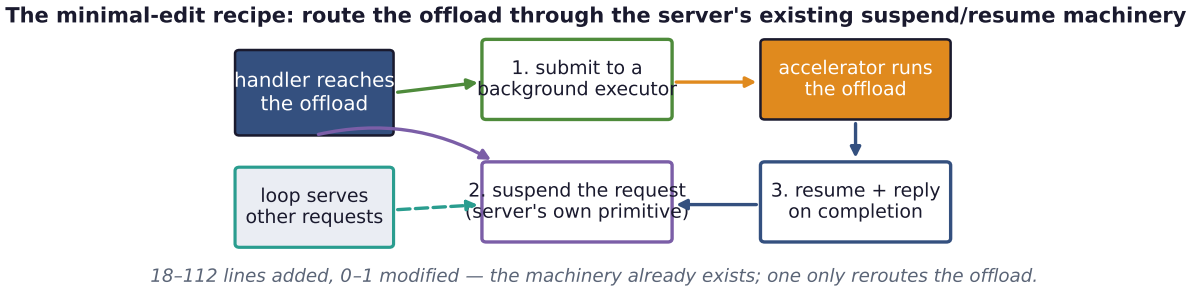

Figure 8. The minimal-edit recipe. At the offload call site, submit the work to a background executor, suspend the request with the server's own deferred-response primitive, and resume it to reply on completion. Meanwhile the server's loop serves other requests. The edit is small because the suspend/resume machinery already exists. One only reroutes the offload through it.

4 Minimal-Edit Overlap

If a server cannot be fiberized from the outside, it can almost always be overlapped from a small edit inside. The reason is that every modern server already contains the machinery to *suspend* a request and *resume* it later. That machinery is what lets it serve many connections at once. To overlap an offload, one only has to route it through that machinery instead of blocking on it. The recipe is uniform across servers:

1. at the offload call site, *submit* the work to a background executor instead of waiting.
2. *suspend* the current request using the server's own deferred-response primitive.
3. *resume* it and send the reply when the offload completes.

We distinguish two quantities: *lines added* (the new integration code) and *existing lines modified* (edits to code already in the server). The second is the invasive, maintenance-costly part. Zero is ideal and is achievable wherever the server exposes a plugin or extension interface. We instantiate the recipe on ten servers spanning every concurrency model, and in every case the change is **18–112 lines added and 0–1 lines modified** (Figure 9).

The integration point follows the concurrency model. Servers with a *plugin or extension API* take the recipe with zero core edits: a Redis Redis Ltd. [2024] module, an nginx F5/NGINX [2024] thread-pool add-on, an Apache Apache Software Foundation [2024] content handler, PostgreSQL PostgreSQL Global Development Group [2024] and MariaDB MariaDB Foundation [2024] user-defined functions, a Node.js OpenJS Foundation [2024] N-API addon over libuv libuv contributors [2024], and a HAProxy HAProxy Technologies [2024a] deployment that streams each request to an external offload *agent* over its native SPOE engine HAProxy Technologies [2024b]. Servers backed by a *language runtime* need *no asynchronous code at all*: a Go The Go Authors [2024] handler's plain blocking `cgo` offload is overlapped by the goroutine scheduler, and a Python `asyncio` Python Software Foundation [2024] service offloads through `run_in_executor` in one line. The lone case touching existing code is memcached memcached [2024], which has no asynchronous-command interface, so we add the park/resume transition to its state machine (70 lines, one modified). memcached is the exception that proves the rule: the line count is dominated by whether the server already exposes a deferred-response primitive, which we formalize next.

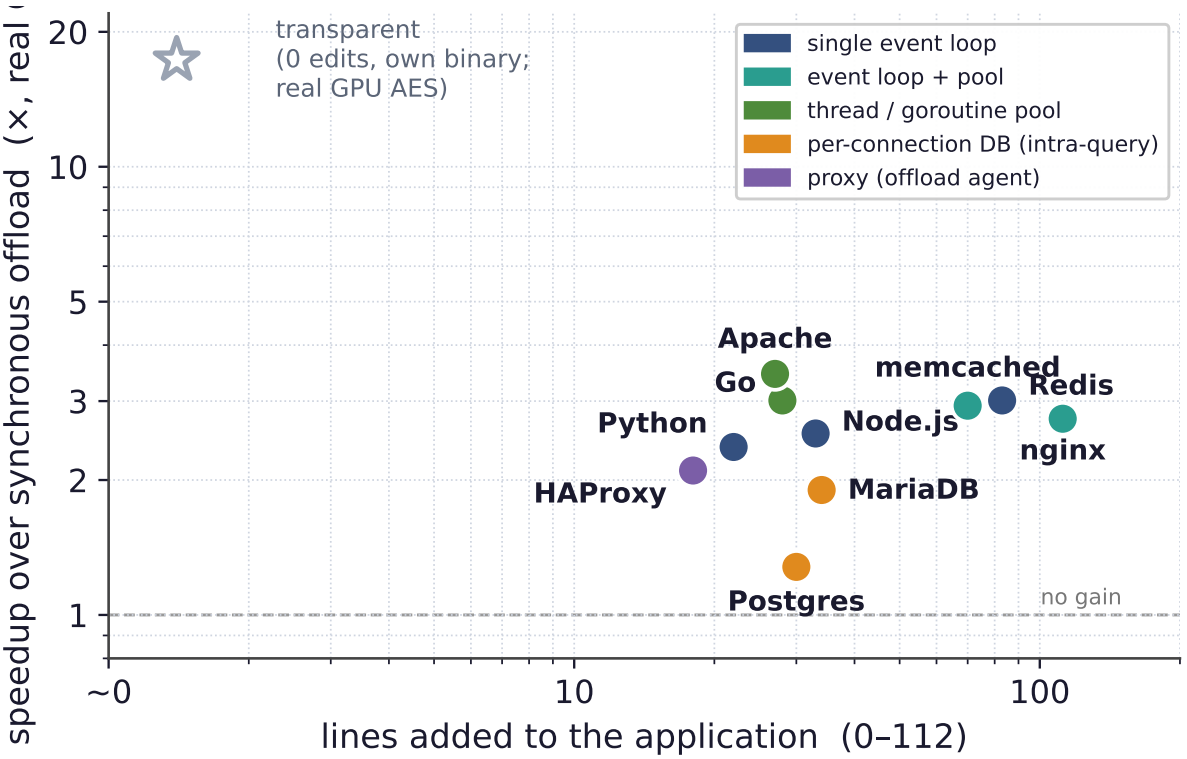

Figure 9. The spectrum (the paper in one plot), all on a *real GPU* with a 1 MiB AES offload, measured on an idle GPU. Each point is an off-the-shelf server. The x-axis is the lines of integration code, the y-axis the real-GPU speedup from overlapping the offload. The few-line edit clusters at 2–3.5×. A real bandwidth-bound GPU offload lets overlap fill the device but not exceed its throughput. The databases sit lower (intra-query pipelining only, as their backends already overlap across connections). HAProxy's proxy reaches 2.1× with a C offload agent (a GIL-bound Python agent gets only 0.9×). The zero-line transparent runtime reaches 17.3× in its niche on a small launch-bound offload (walls on third-party binaries). Color encodes the concurrency model (§5).

5 A Predictive Model of Offload Overlap

The measurements above are not a list of unrelated wins. They follow a pattern. Two properties of a server predict both *how much* overlap helps and *how many lines* it takes: the server's *concurrency model* and the offload's *weight relative to per-request CPU work*.

Concurrency model predicts the regime. Figure 10 lays out the regimes, from *dramatic* to *none*. A *single event loop* is the extreme case—a synchronous offload stalls the whole server, so the few-line fix restores full concurrency and the win grows with offload weight up to the throughput ceiling—while an *event loop with a worker pool* stalls only one loop of several. A *thread or goroutine pool* overlaps offloads *automatically* with zero asynchronous code, and a *proxy* with a native offload engine makes async offload configuration-only. A *process- or thread-per-connection* server already overlaps *across* connections for free (the OS runs the backends concurrently), so its only edit-addressable win is *within* a single query that pipelines a serial loop of offloads.

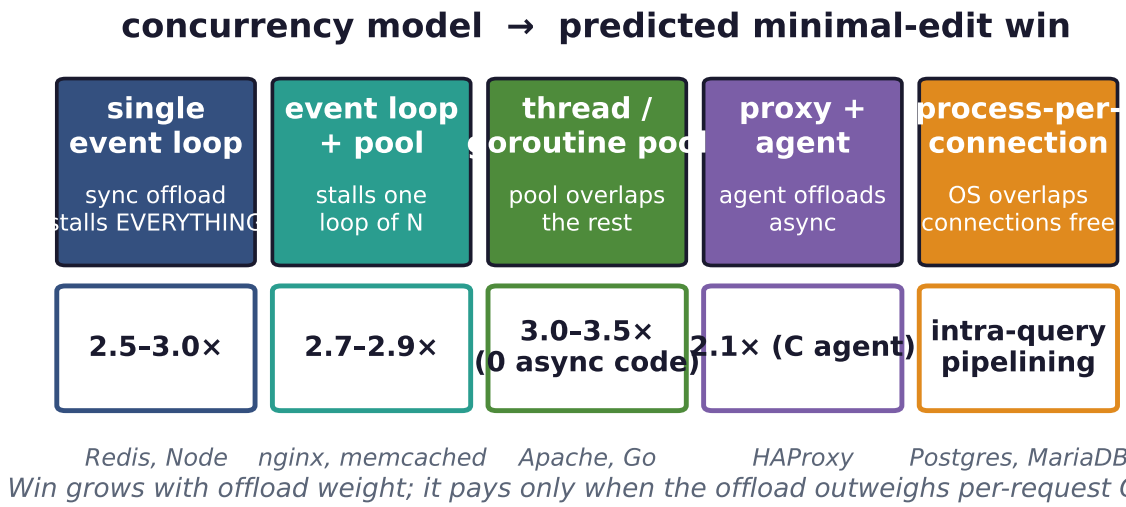

Figure 10. The predictive model, part one: the concurrency model determines what a synchronous offload costs and therefore how much a minimal edit recovers. The win ranges from *dramatic* (a blocked single loop) to *automatic* (a pool that overlaps for free) to *intra-query* (databases that already overlap across connections). Example servers are shown under each regime.

Code cost follows the same axis. The number of lines is governed by whether the server exposes an asynchronous, deferred-response primitive. Plugin and extension interfaces (Redis, nginx, Apache, the databases) and language runtimes (Go, Python) provide one, so the integration is purely additive. No existing lines change. A server without such a primitive forces the developer to add the suspend/resume transition to its request state machine by hand. memcached is our only such case, and it is the only one that modifies an existing line.

Offload weight predicts whether overlap helps at all. The second axis is independent of the server (Figure 11). Overlap reclaims the wait for the offload, so if per-request CPU work is already comparable to the offload there is little to reclaim. A block-size sweep of a real GPU AES offload on a single event loop bears this out: a light block yields essentially no speedup and the win grows monotonically with weight toward the GPU's bandwidth (Figure 11). The rule is simple and bounds applicability: overlap pays exactly when the offload outweighs the per-request CPU work *and* the accelerator is not already saturated—the same condition as Wall 3 of §3, seen from the transparent side.

Together the two axes form a map: given a server's concurrency model and an offload's latency, one can predict before writing any code whether to bother, how large the win will be, and roughly how many lines it will take.

6 Correctness: Overlapping Stateful Offloads

Overlap reorders work, and reordering is dangerous when the work touches shared state. If the post-processing after an offload does a read-modify-write on a global—a counter, a cache entry, a key in a store—overlapping two requests can interleave their sequences and lose updates. On a *stock Redis server* whose module command reads a key, runs a real GPU offload, and writes the incremented value back, unprotected overlap silently loses tens of thousands

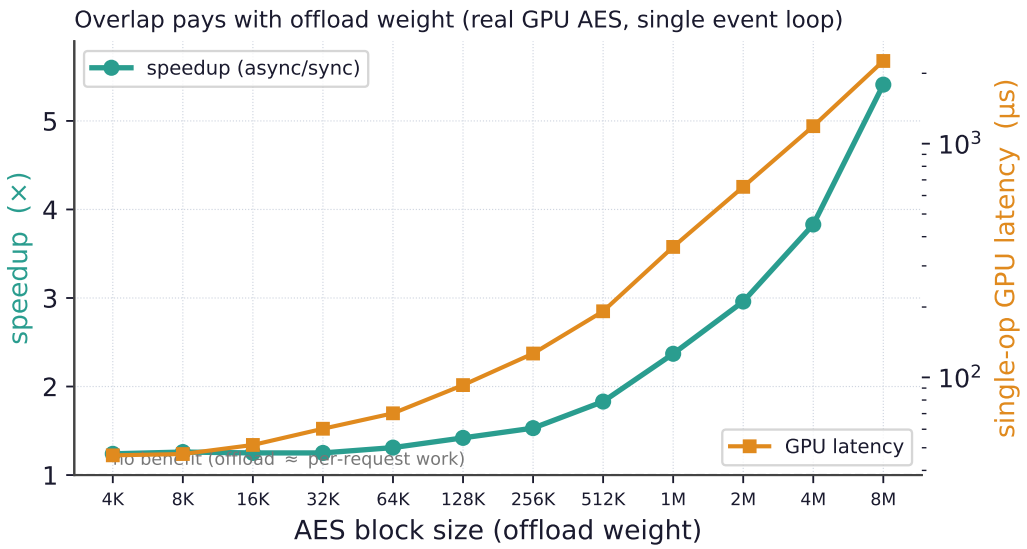

Figure 11. The predictive model, part two (measured, real GPU, idle): overlap pays only when the offload outweighs per-request CPU work. Sweeping a real GPU AES offload by block size (consecutive powers of two, 4 KiB–8 MiB) on a single-event-loop server, the same edit gives no benefit for a light block (1.24× at 4 KiB, 46 µs) and rises to 5.41× at 8 MiB (2.3 ms), approaching the GPU's bandwidth. Right axis: measured single-op GPU latency. This condition also explains the transparent runtime's third wall.

of updates (Figure 12, left). The hazard is subtle precisely because the offload widens the read-modify-write window: the more latency overlap hides, the wider the race it opens.

Two mechanisms restore correctness with no application change. *Per-connection ordering* is free: one fiber per connection serializes a connection's dependent requests automatically. For state shared *across* connections we add a transparent *conflict detector* that write-protects the shared-state pages, snapshots a version clock when a handler parks at its offload, and treats a write fault on a page modified during the park as a conflict; under enforcement it serializes only the conflicting handlers while everyone else stays overlapped. On the stock-Redis workload it flags essentially every conflict and drives lost updates to *zero*, and it keeps overlap where a lock would destroy it—running 1.8× faster than a coarse lock that serializes the very offloads it protects (Figure 12, right). The detector is a property of the overlap mechanism, not of any one integration, so it applies to both ends of the spectrum.

7 Evaluation

Setup. All offloads are real, on real hardware, with no emulated latencies. Experiments run on a server with an NVIDIA RTX PRO 6000 (Blackwell) GPU. The GPU path performs AES OpenSSL Project [2024] on its own CUDA stream, polled by `cudaEventQuery` NVIDIA Corporation [2024], with the block size sweeping from 4 KiB (launch-bound) to 8 MiB (bandwidth-bound). For the latency-bound *remote* class (HSM signatures, post-quantum KEMs, remote inference) we stand up a real TCP signer doing a genuine RSA-2048 signature per request. Event-driven servers are driven by standard load generators (`redis-benchmark`, `ab`). Figure 9 is the cross-server summary; the per-server code cost is in §4.

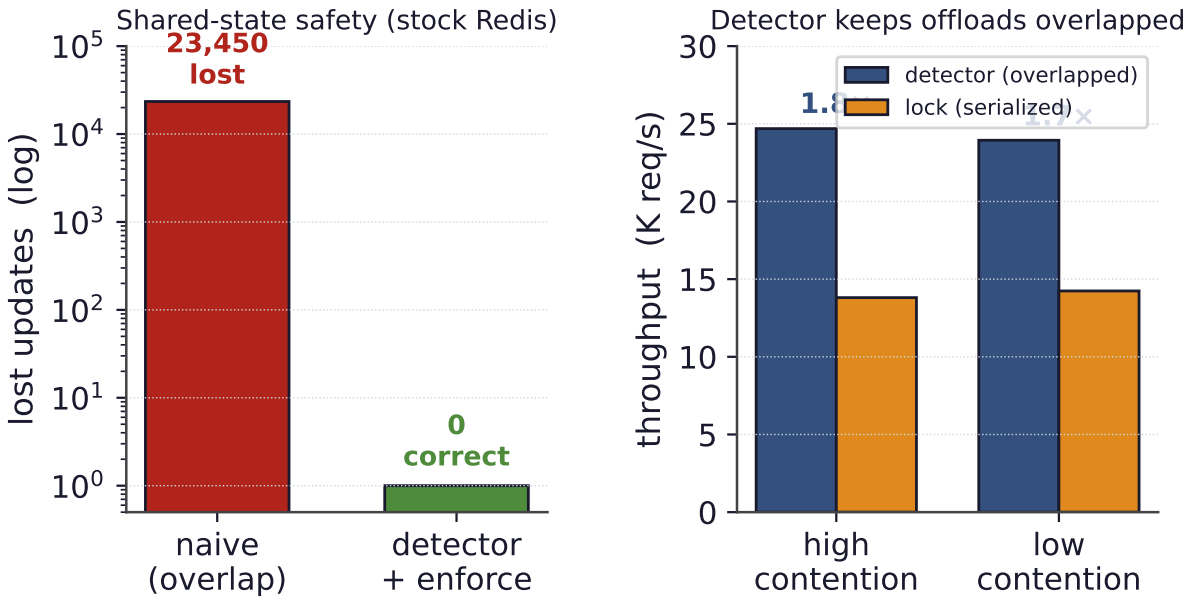

Figure 12. Keeping overlapped stateful offloads correct, measured on a stock Redis server with a real GPU offload. Left: unprotected overlap of a shared read-modify-write loses 23,450 updates. The page-protection detector flags 26,185 conflicts and, under enforcement, reduces lost updates to zero. Right: a coarse lock serializes the very offloads it protects (13.8K req/s), while the detector keeps non-conflicting offloads overlapped (24.7K req/s), 1.8× faster.

Event-driven servers and pools. On a 1 MiB GPU AES offload (realistic bulk crypto, idle GPU) the few-line edit recovers the expected win in every regime with no failed requests: single event loops and event-loop-with-pool servers cluster around 2.5–3×, and thread/goroutine pools (Apache, Go) reach 3.0–3.5× with *no* asynchronous code, since the runtime simply runs other workers while one blocks (Figure 1). The cluster sits at 2–3.5× because a 1 MiB GPU AES offload is bandwidth-bound: overlap fills the device but cannot exceed its throughput. The proxy is the exception that locates the bottleneck: HAProxy's win rides entirely on the offload *agent*, so a GIL-bound Python agent gains nothing while a C `pthread`-pool agent recovers 2.1× on the same offload.

Latency, not just throughput. Overlap is a latency win too. By keeping the CPU busy during offloads, the overlapping path holds both its median and tail (p99) latency low to roughly *four times* the offered load that blocking can sustain before its knee (Figure 13).

Offload weight and the two ceilings. Sweeping the GPU AES offload by block size on a single event loop confirms the weight dependence: no benefit for a light block, rising monotonically toward the GPU's bandwidth for a heavy one (Figure 11; full table in Appendix A). The ceiling is a property of the *device*: overlap recovers the utilization that blocking wastes but cannot exceed device throughput. A *latency-bound* offload lifts that ceiling, because its device is not saturated and many requests can be in flight at once. With our real RSA-2048 remote signer (821 µs round-trip) the same Python server reaches 3.53×—but *not* the order of magnitude a deep queue could in principle give, because the bottleneck simply moves to the server's own overlap capacity (the `asyncio`/GIL executor keeps only ∼7 requests truly in flight). Real

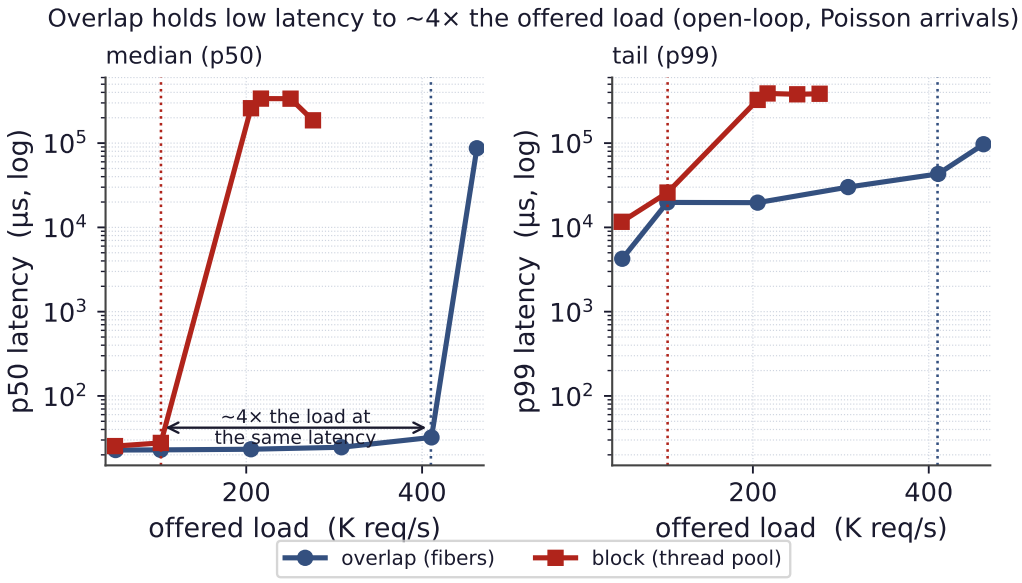

Figure 13. Latency under load (measured, open-loop Poisson arrivals, `runtime/openloop.csv`). Blocking on the offload drives both median (p50, left) and tail (p99, right) latency up at its saturation point near 103 K req/s, while overlapping holds low latency to roughly 4× that offered load (knee near 410 K req/s) before its own. Overlap is a latency win, not only a throughput win.

overlap is thus bounded at *both* ends, by device throughput and by the concurrency the server can express; across our servers and offloads it lands at 1.2–5×.

Databases. A process- or thread-per-connection database already overlaps offloads *across* connections for free (the OS runs the backends concurrently), so the only edit-addressable win is *within* one query that pipelines many offloads. With a launch-bound GPU op this intra-query pipelining is modest (Figure 9), set by the device once it is fed—the same edit, GPU, and per-row work, with the win bounded by the device exactly as the model predicts.

8 Discussion and Limitations

The walls are fundamental, and the win is bounded. The three walls of §3 are properties of where an application places behavior, not gaps in our engineering: raw-syscall synchronization, thread identity in a register, and a workload with no idle CPU each sit *below* or *around* the symbol layer a preloaded library can touch, and they generalize to any storage engine with its own futexes, any managed runtime, and any CPU-bound handler—which is precisely why the minimal-edit path exists. The same idle-CPU condition (Wall 3 and Figure 11) also bounds when overlap pays at all: when per-request CPU rivals the offload there is little to reclaim, and the technique should not be applied.

Detector scope and complementarity. The conflict detector is a targeted safety net for the overlap hazard, not a general transactional memory; shared state outside the monitored segments, or correctness beyond last-writer-wins, would need stronger machinery. Finally, the two ends of the spectrum are a division of labor, not competitors: transparent fiberization serves thread-per-connection servers with no source access, and the minimal-edit recipe serves

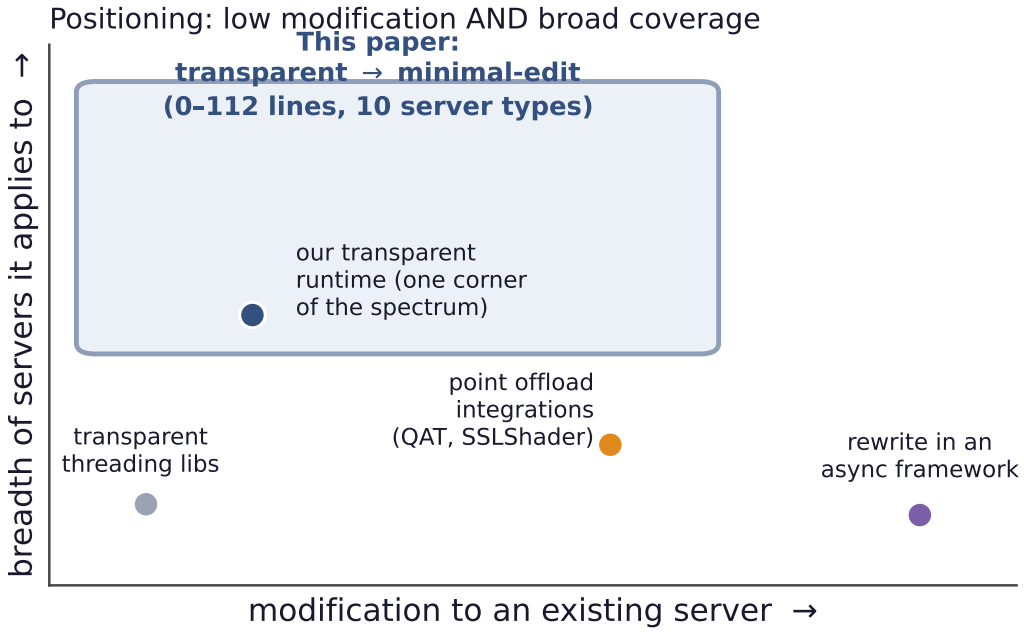

Figure 14. Positioning. Existing approaches to overlapping work are either low-effort but narrow (threading libraries, of which our own transparent runtime is one) or broad but high-effort (point integrations, async-framework rewrites). The transparent-to-minimal-edit spectrum reaches the desirable region: little modification to existing servers, across many server types.

everything else—the event-driven bulk of modern serving best of all.

9 Related Work

Our work sits at an under-occupied point in the design space (Figure 14): low modification to *existing* servers *and* broad coverage across server types. Prior approaches achieve one or the other, but not both. Transparent threading libraries require writing the application in their style, point offload integrations target a single server, and async frameworks assume a rewrite.

Threads, events, and coroutines. Hiding I/O latency cheaply is an old debate, between an event-driven camp that structures the server as a state machine over an event loop at the cost of "stack ripping" (Flash Pai et al. [1999], SEDA Welsh et al. [2001]) and a user-level-thread camp that keeps the synchronous style and yields at blocking points (Capriccio von Behren et al. [2003], State Threads State Threads Library [2009], `libtask` Cox [2005], core-aware Arachne Qin et al. [2018], and language-level threads such as goroutines The Go Authors [2024] and Java virtual threads OpenJDK [2023]). Our fiber runtime is in the second lineage with a fast register-only switch Li et al. [2023], but the contribution is not another threading library: it is to *overlap an accelerator offload* in servers built either way, *quantify the modification cost* across real servers, and map where the transparent form provably stops.

Microsecond-scale systems. A body of work rebuilds the OS or runtime to schedule microsecond-scale tasks efficiently: dataplane operating systems such as IX Belay et al. [2014], work-stealing for tail latency such as ZygOS Prekas et al. [2017], and core-reallocating runtimes such as

Shenango Ousterhout et al. [2019], Caladan Fried et al. [2020], and Demikernel Zhang et al. [2021], typically over kernel-bypass I/O (DPDK DPDK Project [2024], `io_uring` Axboe [2019]). They attack the same killer microsecond problem Barroso et al. [2017] but demand a new stack and rewritten applications. We instead ask how little it costs to overlap one offload inside existing, unmodified servers; the directions are complementary.

Full hardware offload. Another line moves the *entire* datapath onto hardware: KV-Direct Li et al. [2017] runs a key-value store in the NIC, ClickNP Li et al. [2016] compiles network functions to an FPGA, iPipe Liu et al. [2018] offloads application logic onto SmartNICs (and must itself schedule the offloaded tasks' granularity, the dual of our problem), and GPUnet Kim et al. [2014] lets GPU code drive the network. These eliminate the CPU's role and require rebuilding the application around the accelerator. Our target is the opposite and far more common case: the application stays on the CPU and must hide one fine-grained step's latency with almost no change.

Offload integrations and async frameworks. Closer to us, specific systems overlap specific offloads—TLS on GPUs (SSLShader Jang et al. [2011]), QAT engines in nginx's async paths Intel Corporation [2024], and inference servers that batch GPU work (RedisAI RedisAI [2022], Clipper Crankshaw et al. [2017])—but each is a point integration tuned to one server and offload. Async frameworks (libevent Provos and Mathewson [2024], libuv libuv contributors [2024], Seastar ScyllaDB [2024], `async/await`) and proxy offload engines (HAProxy SPOE HAProxy Technologies [2024b], Envoy `ext_proc` Envoy Project [2024]) make overlap the default, but only if the application is written in them from the start. We instead inject overlap into existing servers and give a *predictive model* of the win across the landscape.

Conflict detection. Page-protection dirty-tracking, software distributed shared memory Amza et al. [1996], and transactional memory Herlihy and Moss [1993] all detect or prevent conflicting concurrent writes. Our detector borrows the page-protection technique but is deliberately lightweight and specialized to the one hazard overlap introduces: a read-modify-write reordered across an offload.

Symbol interposition. The fragility of interposing versioned glibc symbols is folklore among practitioners. Our condition-variable finding documents a concrete, reproducible instance and its fix.

10 Conclusion

What should the CPU do while it waits for the accelerator? It should *overlap*: do other useful work while the offload is in flight. The real question is how much an existing server must change, and the answer is a spectrum from zero lines to about a hundred. At the zero-edit end a transparent runtime overlaps an unmodified thread-per-connection binary, but it reaches a narrow niche, hits three architectural walls, and cannot help the event-driven servers with the most to gain. A few lines at the offload site, routed through a server's own concurrency mechanism, work everywhere, recovering 1.2–5$\times$ across ten production servers on real hardware. The win is bounded at both ends—by the accelerator's throughput and by

the server's own overlap capacity—so the order-of-magnitude wins an unbounded model predicts do not survive real hardware, except in the purpose-built transparent runtime (17.3×). A simple model, concurrency model crossed with offload weight, predicts both the speedup and the code cost in advance, and a transparent conflict detector keeps overlap correct when it touches shared state. The result is a practical recipe, and a map, for hiding fine-grained accelerator-offload latency in the servers that run online services today.

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A AES Block-Size Sweep (detailed values)

Table 1 gives the full per-size measurements behind the offload-weight curve of Figure 11, all on a real GPU on an *idle* device (no co-tenant compute), driving a single-event-loop server (Python `asyncio` with a 32-thread executor) at 50 concurrent clients. The AES block size is swept over consecutive powers of two from 4 KiB to 8 MiB. For each we report the measured single-op GPU latency (one offload in flight), the synchronous and asynchronous throughput, and their ratio. The speedup grows monotonically from 1.24× at 4 KiB (launch-bound: the offload is tens of microseconds, on the order of the server's own per-request work, so there is little to overlap) to 5.41× at 8 MiB (the offload is bandwidth-bound at ∼2.3 ms, so overlapping it across requests reclaims most of the wait), up to the GPU's throughput ceiling.

block size	GPU latency (µs)	sync (req/s)	async (req/s)	speedup
4 KiB	46.4	3750	4667	1.24×
8 KiB	47.0	3688	4650	1.26×
16 KiB	51.4	3657	4573	1.25×
32 KiB	60.3	3501	4371	1.25×
64 KiB	70.2	3326	4355	1.31×
128 KiB	92.7	2970	4218	1.42×
256 KiB	126.5	2706	4128	1.53×
512 KiB	191.8	2454	4494	1.83×
1 MiB	361.6	1538	3644	2.37×
2 MiB	653.6	966	2858	2.96×
4 MiB	1188.1	506	1937	3.83×
8 MiB	2258.6	227	1228	5.41×

Table 1. Real-GPU AES block-size sweep on a single-event-loop server (idle GPU). Source: `transparent-runtime/apps/aes_blocksize_py_results.csv`.

B Interposition Obstacles for the Transparent Runtime

Beyond the symbol-versioning hazard of §3 (Figure 5), running stock binaries under the `LD_PRELOAD` fiber runtime surfaced four lesser obstacles. Each was resolved once and is recorded here because any threading-interposing tool will meet them.

- **Alternate I/O entry points.** A server may do its socket I/O through `recv/send/poll` rather than `read/write`, so every blocking entry point the application can reach must be interposed to yield the fiber.
- **Real-thread identity.** `pthread_self` must return the genuine OS thread identity, because the C library's stack-bounds logic relies on it; returning a per-fiber value breaks libc internals.
- **Carrier re-establishment after `fork`.** A daemon that forks drops the carrier thread in the child, which must be re-established before any fiber can run.
- **Priority inversion.** Pinning the carrier to a real-time priority can invert priorities against a lock holder running at normal priority, stalling the carrier.

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